Agave Games — Growth Case Study

Ezgi Ekinci

Question 1 Game Analytics & Lifetime Value

General Methodology — Power Law Retention Model

Player retention is modelled as a Power Law curve fitted to three anchor points (t=1, R_1), (t=7, R_7), (t=30, R_{30}) via log-log linear regression:

$$R(t) \;=\; a \cdot t^{-b}$$

Parameters are derived by:

$$b = -\,\mathrm{SLOPE}\big(\ln R_i,\ \ln t_i\big), \qquad a = \exp\big(\mathrm{INTERCEPT}(\ln R_i,\ \ln t_i)\big)$$

6-Month Lifetime (cumulative active days; install day $D_0 = 1.0$ full day):

$$LT_{180} = 1 + \sum_{t=1}^{180} a \cdot t^{-b}$$

LTV and ROAS:

$$LTV_{\text{paid}} = LT_{180} \times ARPDAU, \qquad LTV_{\text{eff}} = LTV_{\text{paid}} \times (1 + \text{Organic Rate})$$

$$ROAS = \frac{LTV_{\text{eff}}}{CPI}, \qquad \text{Net Profit / Install} = LTV_{\text{eff}} - CPI$$

Input Data

Game	D1	D7	D30	CPI ($)	Organic Rate	ARPDAU ($)
Game A	45%	23%	13%	4.00	0.75	0.30
Game B	42%	18%	8%	3.00	1.20	0.28
Game C	50%	25%	10%	2.50	0.50	0.35
Game D	36%	21%	12%	3.00	0.33	0.26

Table 1: Game input data

Fitted Power Law Parameters

Game	a	b	R^2	Model Quality
Game A	0.4550	0.3640	0.9987	Excellent
Game B	0.4322	0.4847	0.9950	Excellent
Game C	0.5333	0.4668	0.9737	Very Good
Game D	0.3692	0.3205	0.9912	Excellent

Table 2: Log-log regression parameters for $R(t) = a \cdot t^{-b}$

Part (a) — 6-Month Lifetime (LT_{180})

Game	LT_{180}	ARPDAU ($)	LTV_{paid} ($)	Org. Rate	LTV_{eff} ($)
Game A	20.01	0.30	6.00	0.75	10.51
Game B	12.60	0.28	3.53	1.20	7.76
Game C	16.25	0.35	5.69	0.50	8.53
Game D	19.20	0.26	4.99	0.33	6.64

Table 3: 6-month lifetime and effective LTV per game

Key observations:

- **Game A** achieves the highest LT_{180} (20.01 days), driven by strong long-term retention (D30 = 13%, the highest among all games).
- **Game D** ranks second in raw lifetime (19.20) but its low ARPDAU and organic rate suppress effective LTV to the lowest of the four games.
- **Game B**'s high organic rate (1.20) provides a meaningful multiplier, yet its short absolute lifetime (12.60) leaves LTV_{eff} below Games A and C.
- **Game C** benefits from the highest D1 retention (50%) and ARPDAU ($0.35), placing it second in effective LTV despite a moderate LT_{180}.

Part (b) — Most Profitable Game at 6 Months

Assumption

A fixed UA budget of $10,000 is applied to all games. Because CPI varies across games, paid install volume differs:

$$\text{Paid Installs} = \frac{\text{Budget}}{\text{CPI}}, \qquad \text{Total Net Profit} = \text{Paid Installs} \times (LTV_{\text{eff}} - CPI)$$

Profitability is compared on *total* net profit (not per-install), capturing the combined effect of unit economics and acquisition volume.

Game	CPI ($)	LTV_{eff} ($)	Net / Install ($)	ROAS	Paid Installs	Total Revenue ($)	Net Profit ($)
Game A	4.00	10.51	6.51	2.63×	2,500	26,266	16,266
Game B	3.00	7.76	4.76	2.59×	3,333	25,864	15,864
Game C	2.50	8.53	6.03	**3.41**×	4,000	34,133	**24,133**
Game D	3.00	6.64	3.64	2.21×	3,333	22,136	12,136

Table 4: ROAS and profitability at $10,000 UA budget (highlighted row = winner)

Winner: Game C ROAS = **3.41**× Total Net Profit = **$24,133** at a $10,000 UA budget

Although Game A has the highest LT_{180}, its CPI of $4.00 caps volume at 2,500 installs. Game C's lower CPI ($2.50) enables 4,000 installs; combined with the best D1 retention (50%) and highest ARPDAU ($0.35), the result is a $7,867 net profit advantage over Game A.

Part (c) — Daily & Cumulative Breakeven: Game A

$1,000/day UA spend, starting 1 September 2026

Methodology

Daily user flow:

$$\text{Paid users/day} = \frac{\$1{,}000}{4.00} = 250, \qquad \text{Effective users/day} = 250 \times 1.75 = 437.5$$

On campaign day D, cohorts from days $1, 2, \ldots, D$ are all simultaneously active. A cohort installed on day k contributes cumulative retention up to age $(D-k)$. Summing across all cohorts collapses to:

$$\text{Revenue}(D) = \text{Eff. users} \times ARPDAU \times \text{CumRet}(D-1), \quad \text{CumRet}(n) = \sum_{t=0}^{n} R(t)$$

Breakeven conditions:

$$\text{Daily BE: Revenue}(D) \geq \$1{,}000 \qquad \text{Cumulative BE: } \sum_{d=1}^{D} \text{Revenue}(d) \geq D \times \$1{,}000$$

Day	Date	CumRet$(D-1)$	Daily Rev ($)	Net Daily ($)	Cum. Revenue ($)	Cum. Net ($)
1	01 Sep 2026	1.000	131.25	−868.75	131.25	−868.75
7	07 Sep 2026	2.879	377.83	−622.17	1,875.05	−5,124.95
14	14 Sep 2026	4.271	560.54	−439.46	5,278.08	−8,721.92
30	30 Sep 2026	6.683	877.13	−122.87	17,055.66	−12,944.34
38	**08 Oct 2026**	**7.698**	**1,010.38**	**+10.38**	**24,679.94**	−13,320.06
60	30 Oct 2026	10.146	1,331.64	+331.64	50,701.08	−9,298.92
80	**19 Nov 2026**	**12.093**	**1,587.19**	**+587.19**	**80,061.96**	**+61.96**
180	27 Feb 2027	19.943	2,617.54	+1,617.54	293,320.05	+113,320.05

Table 5: Game A breakeven progression at $1,000/day (highlighted rows = breakeven days)

- **Daily Breakeven: Campaign Day 38 — October 8, 2026**
  Daily revenue ($1,010.38) first exceeds the $1,000 daily spend threshold.
- **Cumulative Breakeven: Campaign Day 80 — November 19, 2026**
  Total revenue ($80,062) first exceeds total UA spend ($80,000).

Part (d) — New Daily Breakeven After Scaling to $2,000/day

Methodology

After the first daily breakeven (Day 38, 8 Oct 2026), UA spend doubles to $2,000/day. Two cohort pools are now simultaneously active:

- **Phase 1** (Days 1–38): 437.5 eff. users/day ($1k spend)
- **Phase 2** (Days 39+): 875 eff. users/day ($2k spend)

Revenue on day D (for $D > 38$, with $BE_1 = 38$):

$$\text{Rev}(D) = 437.5 \times ARPDAU \times \Big[\text{CumRet}(D{-}1) - \text{CumRet}(D{-}BE_1{-}1)\Big] + 875 \times ARPDAU \times \text{CumRet}(D{-}BE_1{-}1)$$

New daily BE: first D where $\text{Rev}(D) \geq \$2{,}000$.

Day	Date	Phase	Spend ($)	Daily Rev ($)	Net Daily ($)	Daily BE?
38	08 Oct 2026	1 ($1k)	1,000	1,010.38	+10.38	**YES**
39	09 Oct 2026	2 ($2k)	2,000	1,157.52	−842.48	**NO**
45	15 Oct 2026	2 ($2k)	2,000	1,496.47	−503.53	**NO**
55	25 Oct 2026	2 ($2k)	2,000	1,890.54	−109.46	**NO**
59	**29 Oct 2026**	2 ($2k)	**2,000**	**2,028.23**	**+28.23**	**YES**
70	09 Nov 2026	2 ($2k)	2,000	2,374.34	+374.34	**YES**

Table 6: Game A Phase 2 breakeven ($2,000/day; highlighted row = new daily BE)

New Daily Breakeven at $2,000/day: Campaign Day 59 — October 29, 2026

After doubling UA spend on Day 39 (Oct 9), it takes an additional 21 days for combined Phase 1 + Phase 2 revenue to reach the new $2,000/day threshold. The ramp is faster than the original 38-day Phase 1 ramp because Phase 1 cohorts are already mature and contributing significant daily revenue, so Phase 2 only needs to close the incremental $1,000/day gap.

Question 2 Creative Performance — Wonder Link

Analytical Framework

Creative performance is evaluated on two interconnected dimensions:

CPI mechanism — Ad auction platforms (Meta, TikTok, Google UAC) price impressions based on predicted engagement. Higher CTR (click-through rate) earns lower CPM (cost per thousand impressions), which reduces CPI:

$$CPI \approx \frac{CPM}{CTR \times \text{Install Rate}}$$

A stronger hook (greater thumb-stopping power in the first 1–3 seconds) raises CTR, which lowers CPM and therefore lowers CPI. This is the structural mechanic of real-time bidding auctions, not a speculative claim.

Retention mechanism — A creative sets user expectations before install. Authentic gameplay attracts a matched audience; the user finds what they expected and stays. Fabricated or exaggerated mechanics attract a mismatched audience; the user encounters cognitive dissonance and uninstalls immediately, inflating effective CPI and destroying LTV. These two forces are typically in tension:

$$\textit{Lower CPI} \longleftrightarrow \textit{Aggressive / misleading hook} \longleftrightarrow \textit{Lower retention}$$

The optimal scaling creative balances both — authentic enough to protect retention, compelling enough to lower CPI.

Five evaluation criteria

1. **Hook (0–3 s):** Psychological lever; thumb-stopping power
2. **Gameplay authenticity:** Does the ad accurately represent the core loop?
3. **Target audience breadth:** Wide (high volume, fit risk) vs. narrow (low volume, good fit)
4. **CTA:** Is the call-to-action clear and action-driving?
5. **Visual satisfaction:** Does it tease the real reward of the game?

Ad 1 — Positive Dopamine Hook (30 s)

Criterion	Observation
Hook (0–3 s)	Begins immediately with authentic Wonder Link gameplay: coloured balls and breakable crates. A 3D animated cartoon hand creates chains with colored balls and breaks crates. In-game UI is visible: in the top a progress bar and a glowing power-up booster. No text overlay.
Hook mechanism	Positive dopamine / reward cascade: as gameplay progresses the progress bar fills, triggering a first booster; a subsequent long chain triggers a second booster; the two boosters cascade and clear the entire board. This is a high-dopamine "satisfying completion" moment.
Gameplay authenticity	✓ Authentic. Mechanics (colour-matching, chain connections, booster cascade) match the real game precisely. User expectation is accurately set.
Target audience	Narrow: reaches casual puzzle players who already appreciate this mechanic. Low risk of mismatched audience; low volume ceiling.
CTA	Wonder Link logo + "Try it yourself!" + green PLAY! button. No hand clicks the button (static end card). Weakest CVR signal of the three.
Visual satisfaction	High. The board-clear booster cascade is the genuine highlight of the real game.

Metric	Direction	Reason
CPI (nominal)	**High**	Positive hook has lower click than fail/challenge hooks
D1–D7 Retention	**Low** risk	Authentic gameplay; user finds what they saw
Effective CPI	Medium	High retention partially compensates for high nominal CPI

Table 7: Ad 1 directional CPI and retention estimates

Ad 2 — Fail Bait Hook (23 s)

Criterion	Observation
Hook (0–3 s)	The board opens fully filled with balls in two colours. A colour-target counter bar at the top displays five colour goals with numeric counts. An uncluttered two-color board and clear objectives may relieve the player's cognitive load.
Hook mechanism	Fail bait / frustration bait: the animated cartoon hand correctly chains most blue balls in large serpentine S-shaped paths, but deliberately leaves one blue ball unconnected. That single isolated ball causes the level to fail. The viewer's psychological reactance activates: "*I would never make that mistake.*" Install intent is driven by the desire to prove competence.
Gameplay authenticity	✓ Authentic. The fail outcome is exaggerated for drama, but not fabricated.
Target audience	Moderately broad: the fail-bait format appeals to casual and mid-core players alike, widening the funnel without sacrificing audience fit.
CTA	Wonder Link logo + "Try it yourself!" + green PLAY! button with an animated hand clicking the button. This gesture priming technique directs eye movement to the CTA and pre-activates the viewer's motor response, measurably increasing CVR.
Visual satisfaction	Medium. Large S-path chains are satisfying to watch, but the fail ending deliberately withholds the completion reward, deferring gratification to the game itself.
Duration advantage	At 23 s this is the shortest of the three ads. Higher video completion rates earn lower effective CPM from platform algorithms, adding a structural CPI benefit.

Metric	Direction	Reason
CPI (nominal)	**Medium**	Fail hook broadens CTR over Ad 1; authentic gameplay limits breadth below Ad 3
D1–D7 Retention	**Low** risk	Real mechanics; user finds a genuine game, not a bait-and-switch
Effective CPI	**Low**	Best nominal CPI among authentic ads, protected by solid retention

Table 8: Ad 2 directional CPI and retention estimates

Ad 3 — Challenge + Frustration Bait (30 s)

Criterion	Observation
Hook (0–3 s)	Pure white background. Large bold text: "**Can you solve it?**" Subtitle: "Connect the dots without crossing the lines." A minimalist black-and-white circular dot-connection puzzle is displayed — a visual language entirely absent from the actual game's interface.
Hook mechanism	Challenge hook + frustration bait: the opening question is reinforced by a real human hand holding a pen making obviously incorrect connections one after another. Unlike Ad 2's cartoon hand, the live-action element amplifies perceived authenticity: viewers feel more confident they could solve it, intensifying psychological reactance and install intent.
Gameplay authenticity	× Fake. The dot-connection puzzle has no counterpart in Wonder Link's actual gameplay.
Target audience	Maximally broad: the generic logic-challenge aesthetic attracts a far wider audience than the game's natural casual-puzzle player base.
CTA	Wonder Link logo + "Try it yourself!" + green PLAY! button. No hand clicks the button (same static format as Ad 1).
Visual satisfaction	Low. No board-clear, no colour explosion; the ad ends on an unresolved challenge.

Metric	Direction	Reason
CPI (nominal)	**Low**	Maximum audience breadth + dual hook (challenge + frustration) = highest CTR
D0–D1 Retention	**High** risk	Cognitive dissonance: user expects dot puzzle, finds colour-ball game
Effective CPI	**High**	Expectation mismatch triggers immediate uninstall

Table 9: Ad 3 directional CPI and retention estimates

Side-by-Side Comparison

Feature	Ad 1	Ad 2	Ad 3
Duration	30 s	23 s	30 s
Hook type	Positive dopamine	Fail bait	Challenge + fail bait
Gameplay authenticity	✓ Authentic	✓ Authentic	× Fake
Hand type	3D cartoon	3D cartoon	Realistic
Hand clicks CTA	No	Yes	No
Estimated CTR rank	Lowest	Medium	Highest
CPI direction	**High**	**Medium**	**Low**
D1 retention risk	**Low**	**Low**	**High**
Effective CPI	Medium	**Low**	Medium–High

Table 10: Comparison summary

Scaling Recommendation

Recommended for scaling: Ad 2

Three structural arguments support this choice:

1. **CPI advantage over Ad 1:** The fail bait hook activates psychological reactance, generating higher CTR than Ad 1's passive positive hook. Combined with the shortest runtime and a gesture-primed CTA (animated hand clicking PLAY!), Ad 2 is structurally positioned to outperform Ad 1 on CPI.
2. **Retention protection over Ad 3:** Ad 2 uses authentic in-game mechanics; the acquired user cohort has accurate expectations. D1 and D7 retention should remain at levels consistent with the game's organic baseline.
3. **Effective CPI is what matters at scale:**

$$CPI_{\text{effective}} = \frac{\text{UA Spend}}{\text{Retained Users (D7)}}$$

Ad 3's low nominal CPI is destroyed by high immediate churn. Ad 2 retains a far larger share of the installed base, making each dollar of UA spend generate meaningful long-term value.

Question 3 Video Concept Development — What The Hex

Game Context & My Insights: Upon analyzing the ad creatives and gameplay trailers of the top 5 performing 'Hex Sort' competitors in the market, I observed that winning creatives share specific structural elements:

1) Starting with a fully packed board

2) Using a "tutorial hand" to guide or misguide the action

3) Displaying locked board sections to trigger a sense of progression and curiosity

Concept 1: The "Satisfying ASMR" Cascade (Positive Hook)



Figure 1: Hexa Crush's CTA screen.

Core Strategy: Capitalise on the viral "satisfying / tidying" aesthetic popular on TikTok. Sorting games thrive on the psychological reward of transitioning from chaos to order, amplified by unlocking new progression areas.

Reference Media Note: You can view the massive board-clearing cascade that inspired this concept (*my gameplay*) directly via this link: `https://drive.google.com/drive/folders/1nTdpoTaf6KTTDHph8lPEpsYUUT537lky?usp=sharing`

- **Visual Hook (0–3 s):** Opens on a massive, deeply chaotic board that is critically full . This creates an immediate sense of tension and claustrophobia.
- **Action & Audio:** A tutorial hand slowly places a single, crucial hex stack into one of the final empty slots. As it lands, a highly satisfying, crisp ASMR sound (e.g., wooden clicks, soft magnetic snaps) begins.
- **The Climax:** That single placement triggers a massive, continuous chain reaction. Colour after colour merges in rapid succession, successfully clearing the board and completing

the level. As the level concludes, the screen shifts focus to the locked sections (padlock symbols) representing the upcoming new levels and challenges.

- **CTA:** The glowing padlocks invite the player to continue the journey, directly pointing to a pulsating "PLAY NOW" button to unlock them. Inspiration can be drawn from the CTA screen of Hexa Crush, shown in Figure 1.
- **Why it reduces CPI?** Adding the "near-failure" tension right before the massive chain-reaction release amplifies the psychological payoff (dopamine hit). Concluding with locked symbols for the next levels triggers a strong sense of curiosity and progression, driving the user to install to unlock the remaining content.

Concept 2: The "Obvious Mistake" Fail Bait (Reactance Hook)

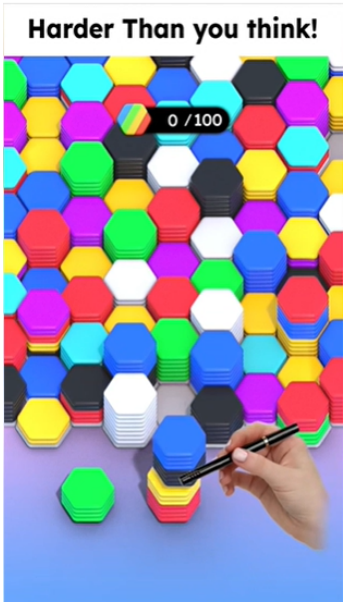


Figure 2: Hexa Crush's hook.

Core Strategy: Trigger psychological reactance. Viewers are highly motivated to correct someone else's blatant error, driving them to click and install the game to prove their own competence.

- **Visual Hook (0–3 s):** A fully filled puzzle board where Text overlay reads: *"Only 5% can unlock this!"*.
- **Action & Audio:**The omnipresent tutorial hand hovers over the board, executing two accurate, successful placements to teach the core mechanics. Then, faced with one incredibly obvious move to clear the board, the hand suddenly hesitates, moves the wrong stack, and fails the level.
- **The Climax:** A harsh buzzer sound plays, the screen turns greyscale for a split second, and a large **"FAIL"** stamp appears.
- **CTA:** A bold "Think you can unlock it?" text appears alongside the animated hand actively clicking the "PLAY!" button (gesture priming).
- **Why it reduces CPI?** People click through the ad out of sheer annoyance at the "incompetent" gameplay preventing the lock from opening. This massive spike in CTR structurally forces the ad platform's algorithm to lower the CPI.

Meta-Gameplay Exclusion

The reference media (app store What The Hex fragman) shows a *"Build Towns"* meta-feature, the video creatives intentionally isolate the core puzzle mechanics. Mixing meta-progression with core problem-solving within a short 30-second ad splits viewer focus and dilutes the psychological hook, which would negatively impact CTR and CPI. Which is why I preferred to leave it out of the ad creatives.